

ABG Interpretation Workshop

Participant Question Booklet - 13 Cases

How this session runs

You have all 13 cases in this booklet. Read them during the 8-minute settling period. Each case then runs as one 15-minute unit in two phases:

Small Group - 8 min - Group faculty	Plenary - 7 min - Moderator
0-3 min: Team interprets independently 3-6 min: Faculty-mediated discussion 6-8 min: Group synthesises answer	0-3 min: Random draw - one group presents 3-6 min: Case-specific Q and A plus post-intervention gas 6-7 min: Take-home pearl

The Unified 10-Step Approach

Step	Tool / key question
1. Clinical history	What does the story alone suggest?
2. Internal validity	$[H^+] = 24 \times PaCO_2 / HCO_3^-$ - does it match pH? Arterial?
3. pH	Acidaemia, alkalaemia, or normal? A normal pH does not mean a normal gas.
4. Primary disorder	Does $PaCO_2$ or HCO_3^- explain the pH direction?
5. Expected compensation	Metabolic acidosis: $PaCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$. Metabolic alkalosis: $PaCO_2 = 0.7 \times HCO_3^- + 21 \pm 2$.
6. Anion gap	$Na^+ - (Cl^- + HCO_3^-)$; add 2.5 per 1 g/dL albumin below 4.0.
7. Delta ratio	$(AG - 12)/(24 - HCO_3^-)$ - only if AG raised.
8. Oxygenation & integration	P/F, OI, CaO_2 , $ScvO_2$, MetHb. Does the story fit the child?
9. SID / Stewart analysis	SID = $Na^+ - Cl^-$ (normal ~36). Low = chloride-driven acidosis. High = chloride-loss alkalosis. Atot = albumin + phosphate. Unmeasured anions = $AG - (lactate + ketones)$.
10. Post-intervention re-evaluation	Has the mechanism changed? Recalculate AG, SID, Atot.

Case 1

Vignette. 10-year-old girl, 2 weeks of polyuria and 3 days of vomiting and abdominal pain. Acidotic breathing present, haemodynamically stable, GCS 11.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.08	14	4	36	132
K	Cl	Albumin	Glucose	b-OHB
4.5	98	4.0	480	5.8

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 98 = 34$	
Atot	Albumin 4.0	
Unmeasured anions	$30 - 5.8 = 24.2$	
Mechanism	-	

Case-specific questions

1. Calculate the corrected serum sodium.
2. What do the glucose and b-OHB values suggest?

Case 2

Vignette. 6-week-old first-born boy with 10 days of progressively forceful, non-bilious vomiting after every feed. Sunken eyes, lethargic but arousable, still hungry after vomiting.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.52	48	38	34	132
K	Cl	Albumin	Glucose	Urine Cl
3.0	82	4.0	90	<15

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 82 = 50$	
Atot	Albumin 4.0	
Unmeasured anions	$AG\ 12 - 0 = 12$	
Mechanism	-	

Case-specific questions

1. What is the significance of the low urine chloride?
2. What is the likely clinical diagnosis?

Case 3

Vignette. 8-month-old girl with 2 days of fever and large-volume loose stools. Sunken eyes, delayed skin turgor, prolonged capillary refill time.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.22	27	11	34	129
K	Cl	Albumin	Glucose	Lactate
3.2	112	3.5	85	1

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$129 - 112 = 17$	
Atot	Albumin 3.5	
Unmeasured anions	$AG\ 6 - 1 = 5$	
Mechanism	-	

Case-specific questions

1. What is the likely cause of the metabolic acidosis in this child?
2. How do the chloride and lactate values help you identify the mechanism?

Case 4

Vignette. 8-month-old girl with 1 day of fever and vomiting. Effortless tachypnoea, haemodynamically stable, lethargic. History of similar episodes in the past.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.24	38	16	38	135
K	Cl	Albumin	Glucose	Lactate
4	107	2	85	3

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$135 - 107 = 28$	
Atot	Albumin 2.0	
Unmeasured anions	$17 - 3 = 14$	
Mechanism	-	

Case-specific questions

1. Calculate the albumin-corrected anion gap.
2. Is there an additional respiratory disorder?
3. What is the significance of recurrent similar episodes in this clinical setting?

Case 5

Vignette. 15-year-old boy with tinnitus, fast breathing, vomiting, and confusion after intentional ingestion.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.42	18	11	108	140
K	Cl	Albumin	Glucose	Lactate
4.2	102	4.1	90	3.2

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$140 - 102 = 38$	
Atot	Albumin 4.1	
Unmeasured anions	$27 - 3.2 = 23.8$	
Mechanism	-	

Case-specific questions

1. A near-normal pH can conceal a mixed disorder. What disorders are present here?
2. What poisoning is suggested by the clinical history and ABG pattern?

Case 6

Vignette. 5-year-old boy, 20 kg, accidental hanging. Post-arrest, ROSC after 15 min CPR, ventilated. PEEP 6, PIP 16, RR 20, I:E 1:2, FiO₂ 100%, TV 80 mL, MAP 17.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
6.88	65	12	70	138
K	Cl	Albumin	Glucose	Lactate
6.2	98	4.1	210	3.2

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 98 = 40$	
Atot	Albumin 4.1	
Unmeasured anions	$28 - 3.2 = 24.8$	
Mechanism	-	

Case-specific questions

1. Identify all components of the mixed acid-base disorder.
2. Calculate the P/F ratio and oxygenation index using the available data.
3. How would you grade the severity of oxygenation failure?

Case 7

Vignette. 4-year-old boy with septic shock and severe anaemia. FiO2 40%, SpO2 99%, ScvO2 42%.

Initial gas

pH	PaCO2	HCO3	PaO2	Na
7.25	26	11	110	138
K	Cl	Albumin	Hb	Lactate
4.3	106	4.1	4.5	7

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 106 = 32$	
Atot	Albumin 4.1	
Unmeasured anions	$21 - 7 = 14$	
Mechanism	-	

Case-specific questions

1. Calculate the P/F ratio.
2. How do you interpret the low ScvO2 despite a normal SpO2?
3. What do the severe anaemia and elevated lactate indicate about oxygen delivery and extraction?

Case 8

Vignette. 10-year-old boy with cyanosis, headache, tachycardia, and lethargy after receiving topical benzocaine. Blood sample drawn looks chocolate-brown. FiO₂ 90%, SpO₂ 85%, SO₂ 65%.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.36	38	21	550	138
K	Cl	Albumin	MetHb	Lactate
4.3	105	4.1	35%	2

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 105 = 33$	
Atot	Albumin 4.1	
Unmeasured anions	$12 - 2 = 10$	
Mechanism	-	

Case-specific questions

1. Calculate the P/F ratio.
2. How do you explain the discrepancy between the very high PaO₂, SpO₂, and SO₂?
3. What is the significance of the chocolate-brown blood and MetHb 35%?
4. What is the most likely diagnosis?

Case 9

Vignette. Baby A: 4 days old, fed well until day 2, now lethargic and tachypnoeic. Afebrile, clear chest. On antibiotics for sepsis. Baby B: 7 days old, vomiting, lethargic, deep rapid breathing. Platelets 60,000/uL, urine ketones 3+.

Initial gas

A pH	A PaCO ₂	A HCO ₃	A SBE	A Na/Cl
7.51	26	20	-1	138/104
A Glucose	A Lactate	B pH	B PaCO ₂	B HCO ₃
85	1.8	7.14	18	6
B SBE	B Na/Cl	B Glucose	B Lactate	
-21	136/100	45	3.0	

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
A SID	$138 - 104 = 34$	
A Unmeasured	$14 - 1.8 = 12.2$	
B SID	$136 - 100 = 36$	
B Unmeasured	$30 - 3 = 27$	

Case-specific questions

1. Name the primary disorder in each baby. Is compensation appropriate in each?
2. Which features does sepsis alone fail to explain?
3. Baby B has an anion gap of 30 but lactate is only 3. What fills the rest?
4. Intervene: what one test do you send in both, how must the sample be handled, and what do you do while waiting?

Case 10

Vignette. 3-year-old, 5 days after bloody diarrhoea, now oliguric and pale. Hb 6.2 g/dL, platelets 40,000/uL, creatinine 3.8 mg/dL.

Initial gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.22	25	10	-16	131
K	Cl	Albumin	Phosphate	iCa
6.8	104	3.0	9.5	0.92

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$131 - 104 = 27$	
Atot	Albumin 3.0 (alkalinising), phosphate 9.5 (markedly acidifying)	
Unmeasured anions	$19.5 - 0 = 19.5$	
Mechanism	-	

Case-specific questions

1. Is compensation appropriate? What are the anion gap and the albumin-corrected anion gap?
2. Calculate the delta ratio. What does it tell you?
3. Using SID and Atot, name the three separate forces acidifying this child.
4. Intervene: which two sidekick values change your next 10 minutes? The team asks for bicarbonate - what do you say?

Case 11

Vignette. 6-year-old with hepatitis A, grade II encephalopathy. INR 4.2, ammonia 180 umol/L.

Initial gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.49	28	21	-1	132
K	Cl	Albumin	Lactate	Glucose
3.0	98	2.2	5.5	45

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 98 = 34$	
Atot	Albumin 2.2	
Unmeasured anions	$17.5 - 5.5 = 12$	
Mechanism	-	

Case-specific questions

1. What is the primary disorder, and why is this child alkaline?
2. Calculate the anion gap, then correct it for albumin. What changes?
3. The base excess is -1, which looks normal. Why is that misleading? How many processes are running at once?
4. Intervene: which two sidekicks need treating in the next few minutes, and why does one of them worsen the encephalopathy?

Case 12

Vignette. 18-month-old, previously well, fasted overnight after gastroenteritis and presented with a seizure. Liver palpable 3 cm. ALT 180 U/L, ammonia 110 umol/L, free fatty acids raised.

Initial gas

Glucose	pH	PaCO ₂	HCO ₃	SBE
30	7.36	36	20	-4
Na/Cl 138/105	Lactate	b-OHB		
	2.0	0.3		

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 105 = 33$	
Atot	Albumin not given	
Unmeasured anions	$13 - 2 = 11$	
Mechanism	-	

Case-specific questions

1. The gas looks almost normal. Why is that the most worrying finding?
2. What should a fasting child gas and ketones look like?
3. Free fatty acids are high but ketones are low. What does that mean? How would hyperinsulinism differ?
4. Intervene: list the critical sample. When exactly must it be drawn, and what do you give immediately afterwards?

Case 13

Vignette. 2-week-old boy with 3 days of vomiting and poor feeding, now mottled and lethargic. Genitalia look normal. Weight below birth weight.

Initial gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.27	28	12.5	-13	122
K	Cl	Glucose		
7.8	98	40		

Systematic interpretation

Step	Your answer
1. Clinical history	
2. Internal validity	
3. pH	
4. Primary disorder	
5. Expected compensation	
6. Anion gap	
7. Delta ratio	
8. Oxygenation & integration	

SID / Stewart analysis

Tool	Value	Interpretation
SID	$122 - 98 = 24$	
Atot	Albumin not given	
Unmeasured anions	$11.5 - 0 = 11.5$	
Mechanism	-	

Case-specific questions

1. Classify the acid-base disorder and check compensation. Is this a HAGMA or a NAGMA?
2. Three sidekick values make the diagnosis before any specialised test. Which, and what is it?
3. A vomiting 2-week-old with pyloric stenosis would show which gas instead?
4. Intervene: what are your first three actions, and which sample must precede them?